

Name: solutions (please print)

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ECE 2202 – Quiz 4

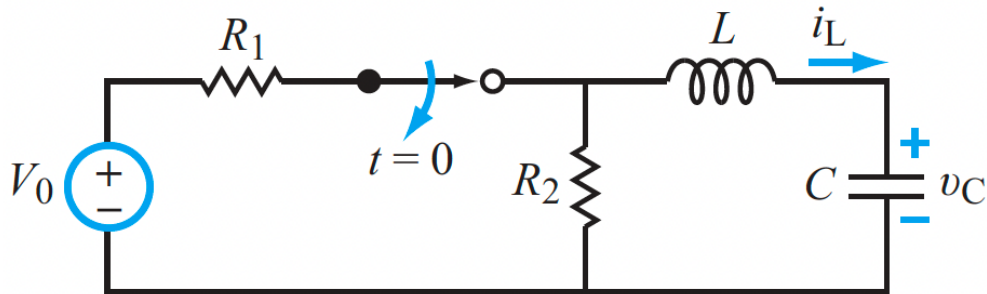
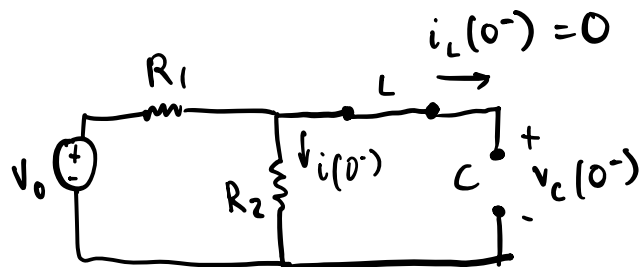
October 23, 2025

1. Show all work necessary to complete the problem. Use additional sheets of paper as needed.
2. Show all units in solutions and figures. Units in the quiz will be included between square brackets.
3. You will have 40 minutes to work on the quiz.
4. Write your ID number on every page that you show your work.

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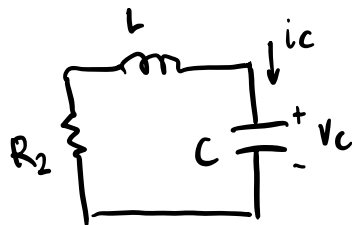
Problem 1 (10 pts)

In the circuit below, $V_0 = 21$ [V], $R_1 = 0.6$ [Ω], $R_2 = 6.3$ [Ω], and $L = 0.8$ [H]. What should the value of C be in order for $i_L(t)$ to exhibit a critically damped response? Provide an expression for $i_L(t)$ and plot its waveform for $t \geq 0$.

 $t = 0^-$ 

initial conditions: $i(0^-) = \frac{V_0}{R_1 + R_2} = 3.04$ [A]
 $v_C(0^-) = i(0^-) \cdot R_2 = 19.17$ [V]
 $i_L(0^-) = 0$

(+) correct values
initial conditions

 $t = 0$ 

(+) redraw series RLC

initial conditions:

$$v_C(0) = v_C(0^-) = 19.17$$
 [V]

$$i_C(0) = i_L(0) = i_L(0^-) = 0$$

next page

determine damping parameters:

$$\alpha = \frac{R_2}{2L} = 3.94 \text{ [s}^{-1}\text{]}$$

for critical damping: $\alpha = \omega_0 \Rightarrow 3.94 \text{ [s}^{-1}\text{]} = \frac{1}{\sqrt{LC}}$

$$C = 0.0806 \text{ [F]}$$

(+2) correct value
+ equations for C

general solution for critically damped:

$$v_c(t) = (B_1 + B_2 t) e^{-\alpha t} + \cancel{v_c(\infty)} \rightarrow 0$$

this circuit configuration is source-free!

looking at table 6-2 in the formula sheet: (+1) B_1 value

$$B_1 = v_c(0) = 19.17 \text{ [V]}, \quad B_2 = v_c'(0) + \alpha v_c(0)$$

$$v_c(t) = (19.17 + B_2 t) e^{-3.94t}$$

$$C [B_2 - 3.94(19.17)] = 0$$

to get B_2 : $i_c(0) = C \cdot v_c'(0) = C [B_2 e^{-3.94t} - 3.94(19.17 + B_2 t) e^{-3.94t}] \Big|_{t=0} = 0$

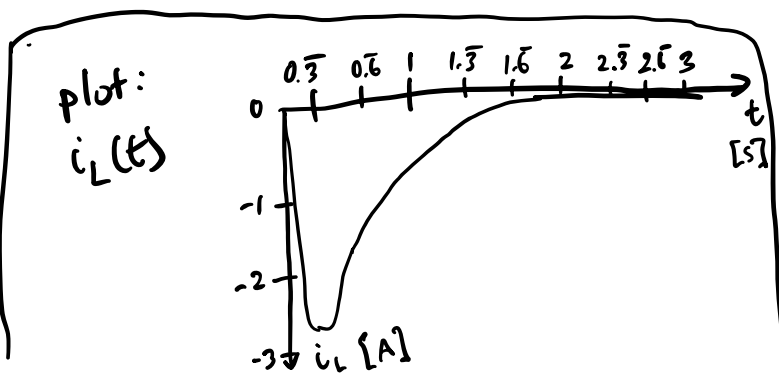
$$B_2 = 75.5$$

$$v_c(t) = (19.17 + 75.5t) e^{-3.94t} \text{ [V]} \quad (+2) B_2 \text{ value and } v_c(t)$$

(+2) final equation

to get $i_L(t)$: $i_L(t) = i_c(t) = C \cdot v_c'(t)$

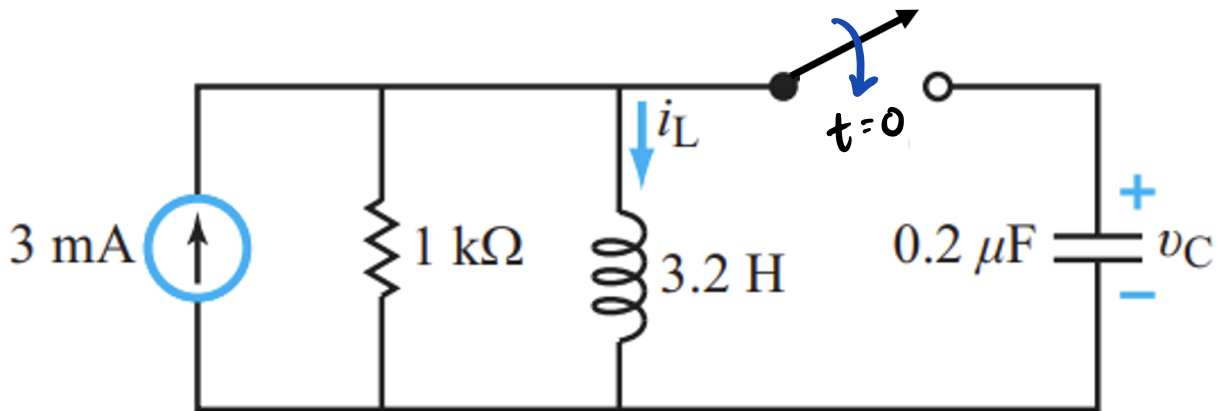
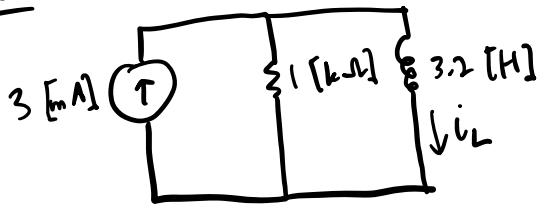
$$= 0.0806 [75.5 e^{-3.94t} - 3.94(19.17 + 75.5t) e^{-3.94t}] = -23.97t e^{-3.94t} \text{ [A]} \quad \text{for } t \geq 0$$



(+1) correct shape
in plot
(no values need
to be labeled)

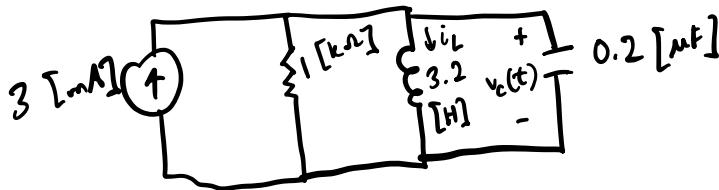
Problem 2 (10 pts)

The switch in the circuit below was closed at $t = 0$. Determine $i_L(t)$ and $v_C(t)$ for $t \geq 0$ [ms]. Assume the capacitor had no charge prior to $t = 0$.

 $t < 0$ 

initial conditions: $i_L(0^-) = 3$ [mA]

(+1) initial condition value

 $t \geq 0$ 

(+1) redraw $t \geq 0$
parallel RLC

damping parameters:

$$\alpha = \frac{1}{2RC} = 2500 \text{ [s}^{-1}\text{]}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 1250 \text{ [rad/s]}$$

$\alpha > \omega_0 \Rightarrow$ overdamped

(+1) damping parameters

next page

general solution for overdamped:

$$i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$$

$$i_L(\infty) = 3 \text{ [mA]}$$

for s_1, s_2 go to table 6-2:

$$s_1 = -4665.1, s_2 = -334.94$$

(+1) roots from table

for $A_1 + A_2$:

$$i_L(0) = A_1 + A_2 + 3 \text{ [mA]} = 3 \text{ [mA]}$$

$$A_1 + A_2 = 0 \quad (1)$$

$$\frac{di_L(0)}{dt} = -4665.1 A_1 - 334.94 A_2 = ?$$

$$v_c(t) = L \cdot \frac{di_L(t)}{dt} \Rightarrow \frac{v_c(t)}{L} = \frac{di_L(t)}{dt} \Rightarrow \frac{v_c(0)}{L} = \frac{0 \text{ [V]}}{3.2 \text{ [H]}} = 0$$

$$\frac{di_L(0)}{dt} = 0$$

$$-4665.1 A_1 - 334.94 A_2 = 0 \quad (2)$$

$$(1) A_1 + A_2 = 0 \Rightarrow A_1 = -A_2$$

$$(2) -4665.1 A_1 - 334.94 A_2 = 0$$

$$A_1 = A_2 = 0$$

(+4) solving for $A_1 + A_2$
with correct equations
+ values

$$i_L(t) = 3 \text{ [mA]}$$

$$v_c(t) = 0 \text{ [V]}$$

(+2) functions
for $i_L(t)$
+ $v_c(t)$

$$i = \frac{dq}{dt} \quad v = \frac{dw}{dq} \quad p = \frac{dw}{dt} = \frac{dw}{dq} \frac{dq}{dt} = iv$$

$$i = C \frac{dv}{dt} \quad v = L \frac{di}{dt}$$

$$v = iR$$

$$R_1 || R_2 = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 R_2}{R_1 + R_2}$$

$$p = Cv \frac{dv}{dt} \quad p = Li \frac{di}{dt}$$

$$p = iv = i^2 R = \frac{v^2}{R}$$

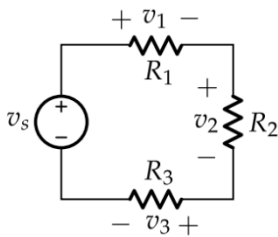
$$w = \frac{1}{2} Li^2 \quad w = \frac{1}{2} Cv^2$$

$$R = \frac{v}{i}$$

$$v_C(t) = v_C(\infty) + [v_C(T_0) - v_C(\infty)] e^{-(t-T_0)/\tau}$$

$$i_L(t) = i_L(\infty) + [i_L(T_0) - i_L(\infty)] e^{-(t-T_0)/\tau}$$

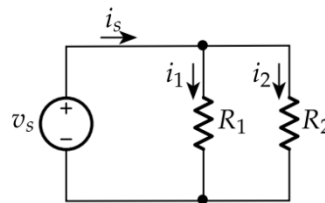
(Switch action at $t = T_0$)



$$v_1 = \frac{R_1}{R_1 + R_2 + R_3} v_s$$

$$v_2 = \frac{R_2}{R_1 + R_2 + R_3} v_s$$

$$v_3 = \frac{R_3}{R_1 + R_2 + R_3} v_s$$



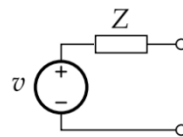
$$i_1 = \frac{R_2}{R_1 + R_2} i_s$$

$$i_2 = \frac{R_1}{R_1 + R_2} i_s$$

$$v_B = \underbrace{V_B}_{\text{DC}} + \underbrace{v_b}_{\text{AC}}$$

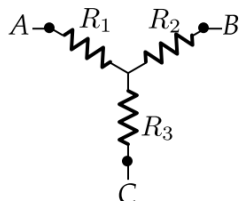
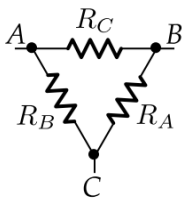
$\bar{V}_b \rightarrow$ Phasor notation

$$\text{CMRR} = 20 \log_{10} \frac{|A_d|}{|A_{cm}|}$$



$$v = i \cdot Z$$

$$i = \frac{v}{Z}$$



$$R_1 = \frac{R_B R_C}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_A R_B}{R_A + R_B + R_C}$$

$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$

Table 6-2: General solution for second-order circuits for $t \geq 0$.

$x(t)$ = unknown variable (voltage or current)	
Differential equation:	$x'' + ax' + bx = c$
Initial conditions:	$x(0)$ and $x'(0)$
Final condition:	$x(\infty) = \frac{c}{b}$
$\alpha = \frac{a}{2}$	$\omega_0 = \sqrt{b}$
Overdamped Response $\alpha > \omega_0$	
$x(t) = [A_1 e^{s_1 t} + A_2 e^{s_2 t} + x(\infty)] u(t)$	
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$
$A_1 = \frac{x'(0) - s_2[x(0) - x(\infty)]}{s_1 - s_2}$	$A_2 = -\left[\frac{x'(0) - s_1[x(0) - x(\infty)]}{s_1 - s_2} \right]$
Critically Damped $\alpha = \omega_0$	
$x(t) = [(B_1 + B_2 t)e^{-\alpha t} + x(\infty)] u(t)$	
$B_1 = x(0) - x(\infty)$	$B_2 = x'(0) + \alpha[x(0) - x(\infty)]$
Underdamped $\alpha < \omega_0$	
$x(t) = [(D_1 \cos \omega_d t + D_2 \sin \omega_d t)e^{-\alpha t} + x(\infty)] u(t)$	
$D_1 = x(0) - x(\infty)$	$D_2 = \frac{x'(0) + \alpha[x(0) - x(\infty)]}{\omega_d}$
$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$	
Auxiliary Relations	
$\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \frac{1}{\sqrt{LC}}$
	$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

$$\vec{S} = \overrightarrow{V_{rms}} \overleftarrow{I_{rms}^*} = |\overrightarrow{I_{rms}}|^2 \vec{Z} = P + jQ$$

$$P = \frac{V_m I_m}{2} \cos \theta$$

$$Q = \frac{V_m I_m}{2} \sin \theta$$

$$pf = \cos \theta$$

$$rf = \sin \theta$$